

FOR 501 Course Outline

FOR 501 — Forestry Sector Modelling

Course Outline

School	05 Sector Planning — Forestry
Level	500 (practitioner / advanced)
Hours	55 (indicative band 40–55)
Shape	5 modules · 25 sessions · 8 laboratories · 4 assignments · 1 final assessment
Delivery	Instructor-led — online, blended or in person
Prerequisites	EDIS 101 required. GIS 711 QGIS or equivalent spatial literacy strongly recommended. Discounting and net present value at the level of a public-investment appraisal, regression at the level of a logistic model, and Julia at the level of reading and modifying a script.
Assessment	Pass 60 %, with 60 % laboratory competency
Version	1.0 · reviewed 2026-09-05 · 12-month cycle

What this course is

FOR 501 teaches a forestry, forest-economy and natural-capital model estate and the judgement required to say what each of its engines does and does not establish. The estate is 92 Julia modules across 9 layers and 23,837 lines: 25 forest and forest-economy engines, 16 AICTEM adapters, 5 mathematical programmes, a geographic layer and a 36-page presentation tier, running from inventory and carbon through deforestation, degradation, growth and harvest to carbon finance, natural-capital accounting, appraisal and the budget.

Every figure in the course is computed on the real Utopia country build, which contains **no generated records at all** — 0 grade-E tables out of 25. The course teaches what that discipline costs as carefully as what it buys: eight tables no source supports are not built, so three national accounts read zero; two levels are calibrated to FAO's reporting rather than derived; the district split of any national total is an allocation; and a national growing stock of 174,617,539 m³ rests on 0 sampled strata out of 405. That is the first thing the course teaches.

The rule the course is built on. Every number carries a grade and a basis. A national total is quotable with its citation; a district figure is an allocation, not a measurement; and a zero may be the correct answer to a question the sources cannot support. Check the grade before you quote, and quote the parameter hash with the figure.

Who it is for

National forestry authority planners, inventory and monitoring officers; forest economists and natural-resource economists; ministry of environment and ministry of finance sector analysts; national statistical office environmental-accounts staff; REDD+ and carbon-finance teams; and modellers joining a forestry-sector team.

Data and access

No learner touches restricted microdata. No laboratory opens a household record, and no survey identifier or cluster coordinate appears anywhere; the Utopia National Panel Survey enters only as the admin-area $\times$ expenditure- quintile aggregates the country build publishes. Every spatial layer — OpenStreetMap extracts, protected-area and watershed boundaries, WorldPop and CHIRPS rasters — is public infrastructure and is mapped freely.

Four labels carry through every figure and the course tests them constantly: **official** for a quantity a national or international authority reported; **derived** for a national total distributed by a stated method; **configured** for a parameter somebody chose and wrote down; and **absent** for a quantity no loaded source supports, which the system reports as a gap rather than a number. Presenting an absent quantity as a model result is an automatic fail.

Structure

Module	Title	Sessions	Laboratory	Hours
M1	The estate, the data and their grades	01–04	01, 02	9.6
M2	The resource: inventory, biomass and carbon	05–08	03	6.9
M3	Loss: change, deforestation, degradation and fire	09–13	04, 05	11.3
M4	Yield and the forest economy	14–18	06	10.0
M5	Value, decision and the handover	19–25	07, 08	17.2
	Total	25	8	55.0

The 25 sessions and 8 laboratories, in teaching order:

M1 The estate, the data and their grades — Sessions 01–04 · 9.6 h · Laboratory 01, 02

- **01 What this system models** — the nine layers, the twenty-five engines, and the run as the unit of provenance
- **02 The data the estate runs on** — thirteen sources, five grades, zero generated records, and eight tables that are not built

- **03 The allocation, and what a district figure can support** — one national forest area, one hundred and thirty-five districts, and a sensitivity table
- **04 Calibration: what it pins and what it leaves free** — two levels that are FAO's rather than the model's
- *Lab 01 The repository, its grades and its gaps* — load the Utopia build, grade every table, and write the data request the gaps imply
- *Lab 02 The allocation and its sensitivity* — take a national total apart across districts and measure how far each district travels

M2 The resource: inventory, biomass and carbon — Sessions 05–08 · 6.9 h · Laboratory 03

- **05 Forest inventory and the basis column** — basal area, volume equations, strata, and an estate sampled by no plots at all
- **06 Biomass and the six carbon pools** — Chave allometry, root-to-shoot, and the pools that are shares of other pools
- **07 Uncertainty as a first-class object** — an estimate with a range, and the sentence that must travel with it
- **08 Classification, concordance and the forest type** — fourteen national classes, and the international classes they map to
- *Lab 03 Inventory, carbon and the uncertainty note* — an estate sampled by no plots, and a carbon range that is honest

M3 Loss: change, deforestation, degradation and fire — Sessions 09–13 · 11.3 h · Laboratory 04, 05

- **09 Cover change and the reconciliation gap** — gross loss, gross gain, net change, and the difference between tree cover and land use
- **10 Fragmentation, edge and the shape of what remains** — the same hectares, arranged two ways
- **11 Deforestation risk: prediction, not policy effect** — a logit, a calibration, and a driver attribution that does not sum to one
- **12 Woodfuel and charcoal: fuel choice before wood volume** — the largest single draw on the forest, and a household energy question first
- **13 Fire: a hazard model, not a spread model** — a logit on temperature, dryness, fuel and access, and two emission figures
- *Lab 04 Cover change, reconciliation and risk* — gross against net, the transition matrix, and a logit that is not a policy model
- *Lab 05 Woodfuel, degradation and fire* — the draw that dominates, and a hazard layer that is not a forecast

M4 Yield and the forest economy — Sessions 14–18 · 10.0 h · Laboratory 06

- **14 Growth and yield** — one stock equation, two growth forms
- **15 Rotation economics and the Faustmann value** — why it is a land expectation value and not a single-rotation NPV
- **16 Sustainable harvest and the four deductions** — the allowable cut, built by removing things, and defended deduction by deduction
- **17 Timber supply, accessibility and the cost curve** — a supply curve built from the cost distribution of the resource
- **18 Non-timber products and the forest industry** — value beside sustainable offtake, and a chain bounded by the forest
- *Lab 06 Growth, rotation and the allowable cut* — re-derive a shipped increment, find why it is

wrong, and build a cut that can be defended

M5 Value, decision and the handover — Sessions 19–25 · 17.2 h · Laboratory 07, 08

- **19 REDD+ and the reference level** — a stock-adjusted reference, a deduction ladder, and the difference between stock and credit
 - **20 Biodiversity, and why it is not carbon** — priority and threat kept apart, and a protection gap
 - **21 Ecosystem services and the three double counts** — a total economic value that can be audited
 - **22 Watersheds and natural capital** — forest to infiltration to sediment to downstream cost, and an account that balances
 - **23 Restoration, appraisal and the budget** — two mathematical programmes, four appraisal bases, and a frontier
 - **24 Community forestry, households and the sixteen adapters** — effectiveness that has to be earned, and the contracts that leave the system
 - **25 Scenarios, governance and the handover** — twelve scenarios, no winner, and the six checks before quoting
 - *Lab 07 Carbon finance, services and the optimised plan* — stock against credit, a TEV that can be audited, and weights that may be doing the choosing
 - *Lab 08 Appraisal, budget and the handover* — four bases, a frontier, a trade-off table with no winner, and the note that goes with them
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What a graduate can do

1. Locate any model in the estate, name the layer that produced a given number, and trace it back to the run id and parameter hash that made it.
2. Read a repository grade table and say, for any figure, whether it is official, secondary, derived, a configured assumption or generated.
3. Distinguish a zero that is a model result from a zero that is an absent source, and write the data request the second implies.
4. Explain what a national total allocated across districts can and cannot support, and read an allocation-sensitivity table.
5. State what a calibration pins and what it leaves free, and report a calibrated level with the authority it actually carries.
6. Compute basal area, tree and stand volume, and expand plot estimates to a stratum and an estate with the basis recorded.
7. Produce a carbon account across six pools with its uncertainty range, and write the sentence that must accompany any published carbon number.
8. Separate gross loss, gross gain and net change, and reconcile a cover series against its transition matrix.
9. Read a deforestation logit, apply a scenario adjustment visibly, and report driver attribution without implying causation.
10. Model woodfuel demand through fuel choice before volume, and distinguish what a stove lever and a kiln lever each act on.

The 23 formal learning outcomes, and the map from each to the assessment that tests it, are in 12_LMS/learning_outcomes.md.

Assessment

Component	Weight	What it is
Knowledge quizzes	15 %	154 items across 5 modules and six question types
Laboratory exercises	30 %	8 laboratories, equally weighted, marked on model competence, interpretation, honesty about limits and communication
Applied assignments	25 %	The estate note and data request; one engine re-derived; the allowable cut defended; the handover note
Final assessment	30 %	Three hours, open book: the estate and its grades; the resource with its uncertainty; loss and yield; and a handover note for a named recipient

Pass 60 % overall and 60 % laboratory competency. Merit 70 %. Distinction 80 %. One resubmission of a single failed component within eight weeks.

The automatic fails

Regardless of the aggregate mark, any of these in an assessed deliverable fails that deliverable:

1. Quoting a district forest area as a measurement. It is an allocation of one national total across 135 districts, and `allocation_sensitivity` reports how far each district travels under other settings.
2. Reporting a driver-attribution share as a policy effect. The model is explicitly `is_causal = false`; the share is what the prediction loses when the driver is neutralised in the equation, not what a policy would deliver.
3. Presenting forest carbon stock as saleable credit. Only reductions against the reference level, net of leakage and the buffer pool, can be issued, and the issuable figure is two orders of magnitude smaller.
4. Summing the three monetary bases of the natural-capital account. Timber resource rent, carbon at the policy price and capitalised services overlap by construction and are deliberately not summed.
5. Adding marketed timber and NTFP output into a total economic value without a production-boundary adjustment — the most common double count in forest valuation, and the one the double-counting report exists to prevent.
6. Reporting a zero as a model result where the diagnostics record `TABLE_UNSOURCED`. The zero is the absence of a source, and the diagnostic names the source that would change it.
7. Quoting a growing-stock or deforestation level as the model's own finding. Both are calibrated to Utopia's reporting to FAO by construction; the pattern across districts is the model's, the level is not.

8. Publishing an optimised restoration or budget plan without its weights and its weight-sensitivity table.
9. Presenting a fire figure as a forecast. The fire model is a hazard logit with no ignition, spread, weather or suppression simulation.
10. Quoting any figure without the run id and parameter hash that produced it.

What this course does not claim

This course does not certify a learner to publish national forest statistics, and it does not treat the Utopia build as a source of operational district figures. Three national accounts read zero on this build because their sources are absent, and the course teaches the data request rather than the zero. Two levels in the system are FAO's rather than the model's. The plantation growth path on this build carries a verified scale defect that the course re-derives and does not quote. Nothing here is an official forestry statistic.

The engine, and the runs behind every figure

Engine	AICTEM Forestry, Forest Economy and Natural Capital Planning System (FFNPS)
Source Case	D:/EDIP/models/ForestryModelling/AICTEM_FORESTRY Utopia, base year 2020, from the estate's real country build
Repository	csv:data\utp
Data status	CONFIGURED_D+DERIVED_C+NO_SURVEY_COVERAGE+OBSERVED_B-
Committed runs	12, all complete with 0 errors
Baseline run	2e4af1a0-92f2-458b-8b5f-c0eb86c25fbb
Parameter hash	f742870c5f253187

Every figure in every deliverable comes from one of those runs and can be reproduced with:

```
cd D:/EDIP/models/ForestryModelling/AICTEM_FORESTRY
julia --pkgimages=no --project=. scripts/run_scenario.jl <scenario_id>
```